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A topic modeling analysis of classroom management and content views and engagement on #TeacherTok

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ABSTRACT

Many novice and pre-service teachers turn to social media platforms like TikTok for on-demand support. Hashtags such as #TeacherTok have become informal professional learning networks that provide accessible, peer-generated content focused on classroom strategies. Drawing on approximately 1400 TikTok videos, this study combined topic modeling with virality metrics to investigate the prevalence, themes, and engagement of classroom management content tagged with #TeacherTok. Findings identified seven salient themes with varied reach and interaction. Viral content often emphasized quick, attention-grabbing strategies; while these posts offered useful tips, they also presented the potential risk of amplifying oversimplified or problematic material due to algorithm-driven preferences. Moreover, TikTok's curation system may build community and provide positive reinforcement, but findings also revealed gaps where less viral yet critical topics remained underrepresented. This study demonstrates the value of combining computational methods with critical media perspectives to examine large-scale teacher discourse in digital spaces.

There is a growing trend for social media platforms to serve beyond entertainment to being used to support information gathering and community development [1]. This also extends to professional roles including educators (both pre-service and early career teachers) who embrace digital spaces such as TikTok, Instagram, or X/Twitter as a form of community and a repository of resources [2,3]. Of specific interest is how social media subject matter may supplement educators' content knowledge development, particularly in areas underrepresented in teacher preparation programs, such as classroom management and supporting student needs. While digital spaces have the potential to increase teachers' access to resources and support there is the question of what content is present, how much users engage with classroom management topics, and potential implications for teachers' professional development. The purpose of this study was to investigate the prevalence and reach of classroom management topics discussed in the #TeacherTok space.

1. Background

Recent scholarship has pointed to the growing engagement with social media spaces as a supplemental tool for teacher preparation [2–5] or form of on-demand professional development, allowing educators to access timely, targeted content that addresses their evolving needs [6,7]. The ease of access to social media platforms enables education-related content to be readily available and allows users to engage with diverse viewpoints, discover new teaching materials, and build connections with others who face similar educational experiences and goals. Rising costs for teacher education courses and professional development workshops have prompted a shift in how social media is used. On-demand professional development and professional learning networks (PLNs) on platforms can offer teachers at all experience levels equitable access to immediate, practical, and often low-cost support [8]. Social media-based PLNs, such as #TeacherTok on TikTok, are transforming these platforms from mere entertainment hubs into valuable and meaningful spaces for education and teacher development, particularly on topics such as classroom management.

As digital spaces and technologies become more integrated into everyday life and instruction, it is increasingly important to understand which content is available, which forms are most popular, and how this content circulates to and is taken up by teachers. For novice or preservice teachers in particular, the material that surfaces most often may shape their perceptions and applications of educational concepts,

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including classroom management, before they have the professional knowledge and experience to evaluate it critically. Historically, social media content has been largely unregulated where misinformation or counterproductive strategies or suggestions can quickly circulate and present as empirically based. Recent research on social media use in teacher education and professional development has centered on how individual teachers engage with these platforms through surveys, interviews, or content analysis of specific accounts or platforms [9]. However, it is important to also extend these methods and take a broader, quantitative approach to capture the full scope of content available. By examining the extensive landscape of social media spaces, especially around topics like classroom management, the aim in this study was to provide a comprehensive view of what exists and what users are actually consuming.

1.1. TikTok: Growing space for educators

Since 2018, TikTok has been a growing international social media platform, featuring short-form videos of three to ten minutes that enable users to connect virtually with followers and build online communities centered on shared interests, identities, and passions [10–12]. While its initial rise to prominence was largely connected to entertainment (e.g., dance trends, make-up tutorials, etc.), it has recently surpassed Google as a platform to search for information [13]. The sheer volume of content consumption within TikTok is staggering, with over 23.56 million videos uploaded daily, and in 2023, the platform had 1.1 billion users [14]. Engagement metrics such as the number of shares, likes, comments, and views a video receives can be used to measure the reach of content or virality [15]. While these metrics can indicate the types of videos and thematic topics most popular or trending in a digital community, they are more reflective of platform design than of neutral or organic user interest.

Niche and interest-driven sub-communities have grown substantially within TikTok tailored to education-focused conversations. These communities have flourished due to active user participation, engagement, and collaboration, especially among teacher accounts with large followings that share pedagogical content [16]. For instance, #TeacherTok alone has amassed over 1.2 million posts as of July 2025, underscoring its potential as a meaningful platform for professional development, the expansion of professional knowledge, and the sharing of instructional strategies. Scholarship examining #TeacherTok or #TeachTok has considered how teachers use the TikTok platform to inform their practice and instruction [2,17–19], including popular hashtags and the emergence of “edu-influencer” [16] or microcelebrity [19]. Hartung et al., [18] posit that intimate realities of classroom teachers are made visible by education-related TikTok content, specifically influencing how the profession is understood by those training to be teachers. As such, social media content can provide accessibility for close observations of authentic classroom practices by offering a space where professionals share struggles, missteps, and frustrations as sources of relevance, relatability, and learning [18].

Additionally, scholars such as Schroeder et al. [20] have examined how education-centered social media content can promote consumerist ideologies, framing teaching through materialized and aestheticized representations rather than pedagogical substance. In parallel, a growing body of research has begun to explore the potential of education-related content on platforms such as TikTok, with scholars examining how teachers and teacher educators leverage social media to support teacher identity development and pedagogical understanding [21]. This emerging work also highlights a tension within social media platforms including education-focused spaces like #TeacherTok. Although content and creators may foreground instructional topics such as classroom management, the underlying emphasis often prioritizes personal branding, material acquisition, and individual gain over meaningful learning or sustained commitment to the broader educational community [22,20]. This dynamic is largely shaped by the underlying nature and structure of the platforms themselves. Scholars in platform studies argue that these systems operate according to a distinct "social media logic," governed by principles of programmability, popularity, connectivity, and datafication, through which platforms both steer user activity and render it into quantifiable metrics [23]. Relatedly, these algorithms both deliver content while also making consequential judgments about what is relevant to a given user. Through this process, users encounter what Gillespie [24] calls "calculated publics" which are algorithmic versions of their community presented to them as an apparent norm. Essentially, what appear to be metrics measuring popularity ultimately work to amplify platform content and skew visibility, such that algorithmic and economic design shapes what users see and how they engage.

1.2. Classroom management and TikTok

The content focus of data in this study is classroom management which encompasses the practice of building and maintaining a classroom environment that supports the social, emotional, and academic learning of all students [25]. It is a proactive and systematic approach that provides a predictable structure in the classroom to promote learning and centers on the development of positive student-teacher relationships [26]. As a topic in education, it is important because it is associated with positive effects on academic, social-emotional, and motivational outcomes for all students, while also protecting teachers’ well-being by reducing the risk of burnout and leaving the profession altogether [27–31].

1.3. Previous work on classroom management in #TeacherTok

As previously mentioned, #TikTok in general and #TeacherTok, specifically, are frequently utilized by teachers for support and resources [2,32,18,19,33,34]. However, how classroom management is presented in these spaces has only recently been studied. Ura and Jerasa [5], for example, explored the utility of #TeacherTok as a classroom management professional development tool. They found that the quality of videos and presentation of evidence-based practices were limited in classroom management videos on TikTok. Likewise, Jerasa and Ura [4] additionally found that while there were limitations of the classroom management videos found on #TeacherTok, there were also some exceptions that could serve as models for teachers who are able to appropriately vet content. Both studies highlight the potential for #TeacherTok to serve as an informal professional development space that could address gaps in teacher preparation.

1.4. Current study

The purpose of this study was to investigate the prevalence and reach of topics discussed in the #TeacherTok space related to classroom management. Using topic modeling, we analyzed a large data corpus scraped from #TeacherTok on classroom management. Identifying what topics are most in demand can inform our understanding of how the information may be taken up and applied in the classroom setting. We specifically addressed the following research questions (RQ):

RQ1. What are the dominant topics related to classroom management present in #TeacherTok? RQ2. Which topics have the most engagement in the #TeacherTok space?

2. Method

2.1. Data collection

To determine appropriate search terms to build our data corpus, the research team undertook a two-week exploratory walkthrough [35] by creating a TikTok account used exclusively for research purposes. During this period, we intentionally engaged with content to train the algorithm to curate videos related to #TeacherTok and classroom management. Drawing from professional expertise, the research team compiled an initial list of keywords they believed teachers might use when searching for classroom management strategies. These terms were then repeatedly searched over the two-week period, while the team maintained collaborative memos to document emergent themes, frequently used hashtags, and the overall content characteristics of the videos that surfaced.

This walkthrough process enabled the refinement and finalization of a focused list of search terms: *classroom management, behavior management, difficult students, difficult behavior, challenging behavior, class control, disruptive behaviors, student disrespect, behavior consequences, escalating behavior(s),* and *discipline.* For video collection, the team conducted targeted searches using the phrase "#TeacherTok + [selected keyword]," aligning their methodology with the established teacher-centered digital space of #TeachTok.

2.2. Data scraping

Data for this study were collected from TikTok using the AllEars tool (https://en.allears.ai), a platform that monitors and collects social media data. The dataset included videos uploaded to TikTok between August 1, 2024, and March 20, 2025. A total of 1346 videos with transcripts were harvested during this period.

2.3. Analysis

2.3.1. Latent Dirichlet Allocation (LDA) topic modeling

To address RQ1 (*What are the dominant topics related to classroom management present in #TeacherTok?*), we explored the dominant themes embedded in the transcripts extracted from TikTok videos. Latent Dirichlet Allocation (LDA) was used as the core topic modeling method. These transcripts are raw and conversational in nature. They often contained informal expressions, overlapping ideas, and digressive speech patterns, which makes them difficult for simpler clustering techniques. LDA was chosen for its ability to probabilistically assign words to latent topics based on patterns of word co-occurrence across the corpus. This approach is particularly effective for uncovering coherent thematic structures in loosely organized, user-generated text data [36–38].

2.3.2. Data preprocessing and cleaning

A multi-step natural language processing (NLP) pipeline was implemented to clean the text data prior to topic modeling. The goal was to reduce noise and irrelevant words and enhance the semantic quality. First, common sources of textual noise were removed. These included URLs, punctuation, special characters, and numbers using regular expressions. Punctuation and numeric characters often contribute little semantic value and may inflate dimensionality unnecessarily [39]. Second, all text was converted to lowercase to ensure that word variants like "Teacher" and "teacher" were treated uniformly during tokenization [40]. Standardizing the case helps reduce vocabulary sparsity. Third, stopwords were also removed. These included common English terms such as "the," "and," and "in," filtered using Natural Language Toolkit's (NLTK) built-in stopword list.

Additional informal speech artifacts, such as "im" (a contraction of "I'm"), also appeared excessively and were excluded. Stopwords typically carry limited topical meaning and can obscure semantic patterns in topic modeling [41,42]. Lemmatization was then performed using the WordNet Lemmatizer. This technique reduces words to their dictionary base form (e.g., "taught" and "teaching" are mapped to "teach"). This method retains semantic coherence better than stemming [43,44].

After cleaning, the text was tokenized and converted into a document-term matrix using the Bag-of-Words (BoW) model. BoW is a widely used vectorization approach that represents each document by word frequency counts while ignoring grammar and word order [45]. The matrix was constructed using CountVectorizer from scikit-learn with the following parameters: max_df=0.95 to exclude extremely frequent terms that appear in >95% of documents, and min_df=2 to discard rare terms. The resulting data served as the input for LDA topic modeling [36,46,47].

2.3.3. Topic evaluation using coherence

To identify the optimal number of topics, coherence scores were calculated with different topic numbers. Coherence provides a quantitative measure of topic interpretability, with higher scores indicating greater semantic relatedness among topic terms [48]. The final best coherence score, prior to human evaluation, was 0.55 for 9 topics.

2.3.4. Human evaluation

To improve the interpretability of LDA-generated topics, Authors 1 and 2 manually reviewed the initial topics ($n = 9$) from output to confirm alignment with the topic descriptor. To do this, both researchers independently selected 5 randomly selected TikTok videos from each categorical topic (total videos checked, n = 90) to assess thematic coherence [49]. Through collaborative discussion, four overlapping topics were merged into two broader themes and final topic labels were assigned to reflect themes grounded in both machine learning output and close human reading of the text data. This hybrid evaluation approach ensured the reliability of the discourse clusters. We report both the topics that emerged and a summary of the content within each category.

To examine the visibility and resonance of different classroom management themes on TikTok and address RQ2 (*Which topics have the most engagement in the #TeacherTok space?*), we calculated the total views of posts within each topic as well as the forms of user engagement with content. While scholars have highlighted that engagement with social media is multidimensional, including aspects of psychological and emotional experience [50], we were most interested in the behavioral aspects of the data, such as number of times the video was liked, shared, or commented on; however, behavioral engagement with a video is difficult to interpret. For example, what is the comparative magnitude and impact of a like versus a share? Likewise, a "like" does not necessarily equate to a positive response to the video–it is simply a single instance of behavioral (positive, negative, or neutral) engagement. Thus, because we were interested in instances of engagement as a contributing factor for virality [15], we combined the total number of likes, comments, and shares to create a sum score for each video to represent the overall level of engagement with videos in each category.

3. Results

The LDA-based topic model and human review revealed 7 dominant topics in the classroom management discourse on Tiktok, including: 1) Positive Prevention, 2) Student Complaints and Deficits, 3) Teacher Agency & Empowering Action, 4) Extrinsic Rewards and Causes for Emotional Behavior, 5) Environment, Materials, and Community, 6) Quick Fixes/Compliance, and 7) Attention to Developmental Needs.

3.1. Dominant topics

The TeacherTok educator community revealed common topics around classroom management where video content provided not only tips and tricks, but deeper questions that teachers wrestle with every day. The following discussion thematically summarizes the seven topics that emerged from the discourse.

3.1.1. Teacher agency & empowering action

This topic features conversations where teachers emphasize educator autonomy, initiative, and personal empowerment in managing classrooms. Conversations included proactive problem-solving, such as teachers sharing stories of taking matters into their own hands, encouraging colleagues to be solution-oriented and try new strategies rather than just complain. One teacher, for instance, reflected that *"the reason why teachers are struggling with behavior management is because they are not willing to change… the only thing in our classroom that needs to change is us."* Many posts also convey subtle resistance to ineffective school policies such as doing what's best for their students even if it means deviating from prescribed methods. The underlying message is that teachers have professional agency and shouldn't feel powerless if a rule or curriculum isn't working. Some educators offer simple but empowering tips, such as participating in fun school spirit events (like dress-up days) with students, saying it builds rapport and the kids *"love to see it."*

3.1.2. Environment, materials, and community

Educators under this theme emphasized the importance of the physical classroom environment, teaching materials, and community-building in managing behavior. Key patterns included practical tips on setting up classrooms for success. For example, arranging desks or centers in ways that minimize disruptions, or having a defined procedure for transitions. One educator described a method to *"help calm the chaos"* when young kids move from carpet time to table work: they assign each table a color and a fun routine so that students move in an orderly, calm manner.

A number of videos highlighted creative use of materials or games to engage students and maintain order. Teachers share resources like behavior tracking charts, noise level monitors, or interactive games that make following rules fun. For example, one teacher introduced a *"mystery student"* game: each day a random student is secretly chosen and if that student meets behavior expectations, the class gets a reward. This game-like approach uses simple materials (name cards or sticks) and turns good behavior into a class challenge. Educators also noted that when students feel the classroom is a community where everyone's role is valued, they are more likely to follow norms. Thus, involving students in setting up the room or creating class rules together was suggested, which gives students ownership and pride in their environment.

3.1.3. Student complaints and deficits

This topic contains conversations that revolve around frustrations with student behavior or skill deficits. Teachers here often vent about challenges they face with students, from lack of respect and motivation to gaps in knowledge or discipline. However, the tone isn't purely negative; mixed in are constructive perspectives addressing those complaints. For example, common complaints include students being disrespectful, disruptive, or unmotivated. For instance, one teacher lamented that students today don't show the same respect, noting that *"your students respecting you is not the same thing as your students liking you,"* implying some students may like a teacher but still challenge authority. Such posts indicate a concern that classroom decorum and student attitudes have worsened, leaving teachers feeling disrespected.

3.1.4. Quick fixes/compliance

This theme captures quick-fix strategies and compliance techniques that teachers shared for immediate results. These were characterized as bite-sized tips, hacks, and sometimes humorous tricks aimed at getting students to follow rules on the spot. For example, one teacher demonstrated a hack called *"Happy Mail"* with a sweet note to the most well-behaving student. Others talk about simple point systems or games used in the moment to regain control. A notable subtheme was using humor, sarcasm, or drama to swiftly manage behavior. An example described an educator dramatically announcing, *"I'm walking towards the phone… now for the moment we've all been waiting for!"* as a student's cell phone rings. Such a playful approach likely made the student quickly put it away. These anecdotes show teachers creatively drawing students' attention through laughter as a compliance tactic. Many strategies in this topic also mention small-scale reward or penalty methods deployed in real time. Teachers talk about giving or taking away points for good or bad behavior, issuing immediate consequences, or leveraging things like one-minute timers, mystery motivators, etc. The very nature of this topic highlights the resourcefulness of teachers in sharing "quick fix" that addresses the noise, chaos, or off-task behavior, etc.

3.1.5. Extrinsic rewards & causes for emotional behavior

This topic blends two areas of the conversation: the use of extrinsic reward systems in classroom management (i.e., charts and prize boxes) and the exploration of emotional causes of student behavior, such as talking about why kids act out (e.g., trauma, emotional needs, etc.). The unifying thread is an examination of what truly drives behavior and how teachers should respond. For example, one educator explained that a monthly raffle approach *"takes away the color charts, the dojo points, the clip charts"* in favor of a simpler, periodic reward, implying it reduces stress and constant competition. In terms of handling emotional behavior, one teacher advised colleagues, *"when dealing with challenging behavior, try not to lead with emotions."* In other words, don't start by yelling or getting visibly upset, as it can escalate the situation. Overall, the cluster of data shows teachers are collectively questioning traditional reward-heavy discipline and educating themselves about the underlying causes of students' behavior. It reveals a value shift in the educator community toward more nuanced, compassionate management strategies.

3.1.6. Attention to developmental needs

Content grouped under this theme highlights the age-appropriate strategies and expectations, focusing on how students' developmental stage affects classroom management. Teachers emphasized that children are not mini-adults. That is, what works for a teenager is very different from what's appropriate for a 5-year-old. The patterns of advice in this topic include: teaching routines step-by-step with younger students, rather than assuming they'll just "get it." Another example suggests having children *whisper answers into their hands before sharing*, as a way to manage the excited blurting out common in younger classes. Conversation in this theme reflects a value of empathy and developmental understanding.

3.1.7. Positive prevention

This largest cluster of the content includes strategies that prevent classroom issues by focusing on the positive. Instead of reacting punitively to misbehavior, these educators stressed setting students up for success through positivity, encouragement, and proactive planning. Overall, the advice highlighted building a strong teacher-student relationship and a supportive class culture as preventative medicine, ignoring minor provocations rather than constantly fighting every small battle, and nurturing a classroom with healthy values rather than dealing with constant "disciplinary emergencies." In many ways, this theme encourages the embracing of preventative care in education, much like preventative medicine–that is, meeting student needs to prevent behavioral issues.

3.2. Reach of video categories

As shown in Fig. 1, content related to *Positive Prevention* emerged as the most widely viewed topic, garnering a cumulative reach of approximately 2.86 million views, followed by *Student Complaints & Deficits* with 2.3 million views. In contrast, the topic labeled *Attention to Developmental Needs*, which included content with play-based learning and age-appropriate instruction, had the lowest cumulative reach (407,002). These results suggest that topics addressing concrete strategies or visible classroom outcomes tend to receive higher view counts than those focusing on broader developmental philosophy.

Fig. 2 represents engagement, a composite of the total number of likes, comments, and shares associated with each topic. It reveals distinct engagement patterns across classroom management topics on

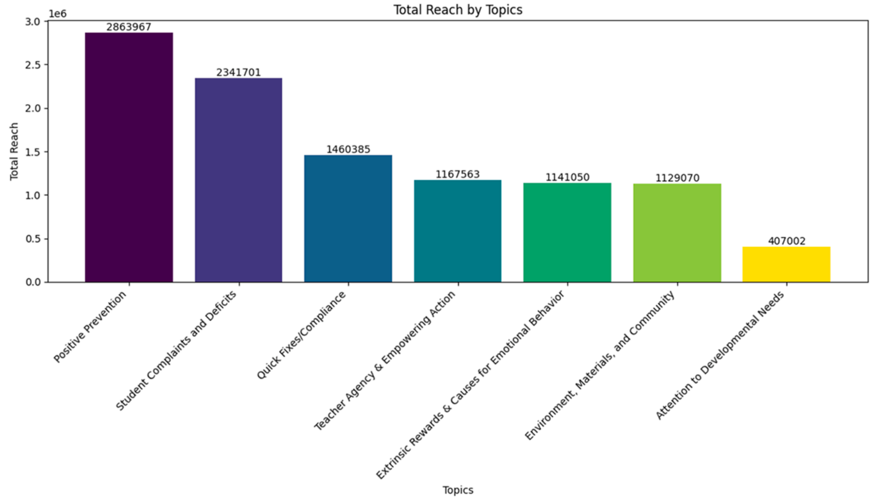

Fig. 1. Total reach of the dominant topics on TikTok.

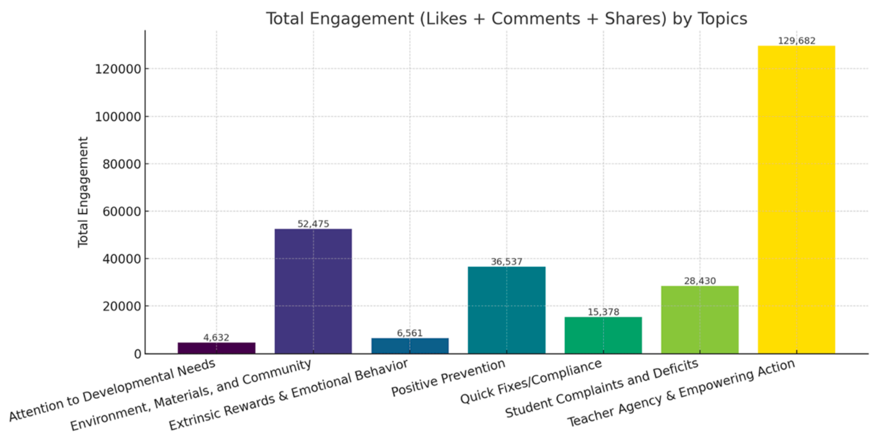

Fig. 2. Engagement with TikTok classroom management posts.

TikTok. Because our analytic interest centered on engagement as a behavioral phenomenon associated with virality, we operationalized engagement as an aggregate measure, combining likes, comments, and shares to capture overall audience interaction [15]. Among all topics, *Teacher Agency & Empowering Action* stands out with the highest engagement (129,682), suggesting that content centered on teacher autonomy and advocacy resonates deeply and is seen as worth engaging. The *Environment, Materials, and Community* category had the second-highest level of engagement. Conversely, *Attention to Developmental Needs* showed the lowest engagement level, pointing to a potential gap in interest or awareness around early developmental issues. Additionally, *Extrinsic Rewards & Causes for Emotional Behavior* drew the second lowest level of engagement (6561), suggesting that emotionally sensitive behavioral content is less popular in the #TeacherTok space. These patterns reflect how different classroom topics vary in emotional impact, perceived utility, and shareability among educators on TikTok.

4. Discussion

Social media spaces, like TikTok, have functioned as largely unregulated platforms, where ineffective strategies can rapidly spread and be framed as evidence-based or grounded in credible research. While TikTok has become a prominent platform for education-related content, there is limited understanding about which topics achieve the greatest visibility and popularity, and how these patterns may shape the overall discourse within the space. The purpose of this study was to identify the dominant topics related to classroom management that are present in #TeacherTok and determine which of those topics have the most engagement. Understanding what is present in such expansive digital spaces like #TeacherTok can provide insight into what content teachers may be exposed to and how this content might shape pedagogical learning.

Our findings suggest seven predominant topics associated with classroom management.

Among these, the research literature supports *Attention to Developmental Needs, Extrinsic Rewards & Causes of Emotional Behavior, Teacher Agency & Empowerment, Environment, Materials, & Community, and Positive Prevention* [51,26,52]. However, the other topics that emerged, including *Student Complaints &Deficits* and *Quick Fixes & Compliance*, reflect some common, but not empirically supported practices. This is concerning because the content within these categories confirms Ura & Jerasa's [5] finding that the focus of many #TeacherTok videos are punitive, compliance-based, and reactive towards student behavior. This suggests that #TeacherTok viewers may encounter contradictory information that is counterproductive to fostering a positive classroom environment, as supported by empirical research.

Surprisingly, there was not a distinctive category focused on teacher-student relationships, which is central to not only classroom management but also classroom pedagogy [53]. While scholarship on teaching and learning prioritizes relationship-building, evidence suggests that the #TeacherTok space does not fully represent empirically grounded practices. We are curious whether this authentically reflects the realities of teaching and classroom management in schools today. Our findings suggest that while relationship building is considered vital, the reality is that other topics, such as quick fixes, overshadow empirically supported approaches (based on engagement metrics).

With regard to the most viewed and most engaged topics, we observed a contrast between the two metrics that warrants closer attention. Within TikTok and other social media platforms, where content is algorithmically curated and brought to users, the topics that are most viewed reflect what users are exposed to and how well searched items align with the platform's calculation and amplification of relevant content [23]. Engagement, however, is a metric of what users deliberately *act upon*, given what has been curated for them. In our findings, *Positive Prevention* was the most viewed topic but ranked only third in engagement, while *Teacher Agency & Empowering Action* received the highest engagement despite being only the fourth most viewed. This distinction suggests a discrepancy between what appears as a norm of the space frontloaded by the algorithm and what users actually choose to act upon [24]. This pattern held as well for *Environment, Materials, and Community* was second in engagement but fifth in number of views. In this example, topics with high engagement but lower views suggest that content actually departs from the expected and prompts deliberate interaction. As such, content within this topic might suggest strategies or materials that viewers want to save for later or share with others. If engagement is understood as one of the few channels through which users exert agency, then it is notable that the most engaged topic centered on teacher autonomy and empowerment, such that users act most readily on content affirming their own capacity and independence. While we cannot determine how viewers ultimately take up these classroom management topics, the divergence between engagement and views suggests that content reflects not only what is viewed but also how viewers respond to what they find most relevant.

While we were able to identify the most watched and most engaged with topics, we also want to highlight the least viewed and engaged with: *Attention to Developmental Needs* and *Extrinsic Rewards & Causes for Emotional Behavior*. There are a few potential explanations for the lower traction of these topics. First, it is plausible that the platform and algorithmic system in TikTok are less conducive to deepen content and user interactions. Much of the viral or trending content within #TeacherTok and TikTok at large focuses on quick, attention-getting, and fast sequences of information. As confirmed by Ura and Jerasa [5], only so much content can be packaged in a byte-sized, short-form video. The metrics and nature of marketing to optimize views, likes, and shares likely reinforce creators to orient content to quick fixes, solutions, and purchasable materials and supports. This may be due to a combination of the lack of discussion around developmental needs and emotional behavior combined with a digital platform that privileges content that is niche, direct, and fast-paced. Thus, the information provided may not give teachers what they need; rather, it may provide what sells. Although classroom management is not often treated as a commodity, videos on TikTok are privileged when they align with the platform's amplification of popular content and the skewing of visibility [23] as well as broader algorithmic preferences for consumption [20]. In addition, students' developmental needs may receive less emphasis in teacher training spaces as they require an understanding of nuanced cognitive aspects of child development. Scholars have noted a tendency to prioritize pedagogy and content knowledge over development and student needs [54,52].

4.1. Limitations & future research

A primary strength of topic modeling is the ability to summarize large amounts of text using an inductive approach, which was appropriate given our research questions. However, some limitations must be mentioned. First, much of TikTok content exists as a multimodal entity, we relied only on text transcriptions of audio and captions present; thus, some of the semiotic meaning-making may have been lost as a result of this form of extraction. Topic modeling foregrounds textual data and, as a result, omits audio, music-based cues, and embodied gestures performed by creators that are central to meaning-making in short-form video. Future research might include a multimodal analytic approach to capture those semiotic tracings and to better account for the sentiment of the content. Second, as Chen et al. [55] point out, there may be some systematic issues with validity of topics, e.g., high predictive power may not necessarily mean representative topics. While we did conduct human evaluation of the topics identified in an effort to ensure validity [48], the number of hand-checked videos at random is a comparatively small sample and potentially leaves room for systematic error.

Regarding our data scraping, we used a third-party platform, AllEars. Due to the application programming interface (API) restrictions placed by TikTok, web-scraping tools are often limited in what they can collect as far as metadata and capturing the full scope of the discourse. While we believe we applied all search terms to elicit the best representation, not having access to the platform's API imposes limitations relative to other platforms with greater public availability. Future research could also consider the confirmation of TikTok data pull by including complementary metadata from other social media platforms, which also include classroom management and educational content.

Finally, measuring and interpreting engagement in social media content is tricky. Vander Schee et al., [50], for example, mentions the importance of attention to not only the instances of engagement, such as likes, comments, and shares, but also points out the psychological and cognitive components involved when interfacing with content. Our focus in this study was on the quantity of engagement, thus, a composite of all three actions was used. However, this in some ways ignores the nuance and potentially unique contribution of each engagement type and its potential effect on how the information contributes to the collective conversation. Future research might focus on this nuance and tease out the differences in impact each type of action (like, share, comment) has on virality, as well as human behavior.

5. Conclusion

Our study suggests potential ways that professional development or teacher education can hook on to the trending topics of classroom management as well as other education-related topics necessary for teachers to be prepared in the classroom. Examining reach and engagement within these spaces allows for a broader understanding of which classroom management topics are most prevalent within #TeacherTok and how such topics are surfaced and encountered by users, who might include teachers seeking professional ideas and perspectives. As a result, this can provide opportunities for professional development and teacher education to address these gaps, since these topics are less represented in such popular social media spaces.

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Ethics statement

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CRediT authorship contribution statement

Sarah Jerasa: Writing – review & editing, Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization. **Sarah K. Ura:** Writing – review & editing, Writing – original draft, Validation, Investigation, Formal analysis, Data curation, Conceptualization. **Md Enamul Kabir:** Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Methodology, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data analyzed in this study were collected from publicly available content on TikTok. Metadata supporting the findings can be provided upon reasonable request to the authors.

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